**Algebra 2 Unit 2A Review** **Name:** ______________________ **Period:** _____

Given the following equations, provide all information and graph the function.

1. Equation: $f(x) = 5|x+1| - 9$

Domain (interval): $(-\infty, \infty)$ Range (interval): $[-9, \infty)$

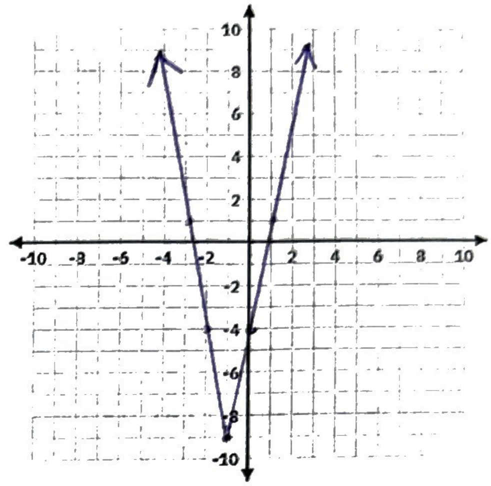
x-intercept(s): $(-2.75, 0)\ (1.75, 0)$ Y-intercept: $(0, -4)$

Vertex: $(-1, -9)$ Axis of Symmetry: $x = -1$

Max: none OR Min: $(-1, -9)$

Function Notation Description: $5f(x+1) - 9$

Verbal Description: VS of 5, Left 1, Down 9



2. Equation: $f(x) = -2|x-3| + 6$

Domain (interval): $(-\infty, \infty)$ Range (interval): $(-\infty, 6]$

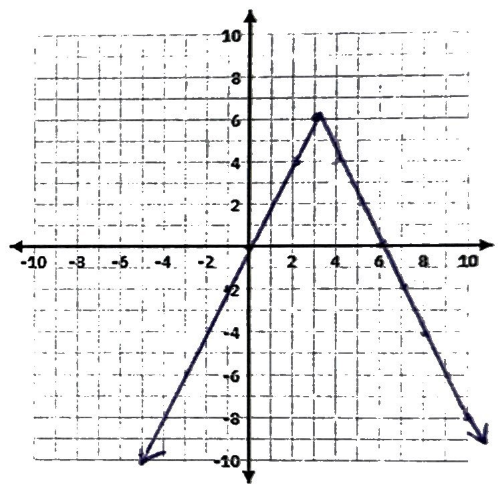
x-intercept(s): $(0,0)\ (6,0)$ Y-intercept: $(0,0)$

Vertex: $(3, 6)$ Axis of Symmetry: $x = 3$

Max: $(3, 6)$ OR Min: none

Function Notation Description: $-2f(x-3)+6$

Verbal Description: Reflect over x, VS by 2, Right 3, Up 6



3. Equation: $f(x) = -\frac{1}{2}|x-4| + 3$

Domain (interval): $(-\infty, \infty)$ Range (interval): $(-\infty, 3]$

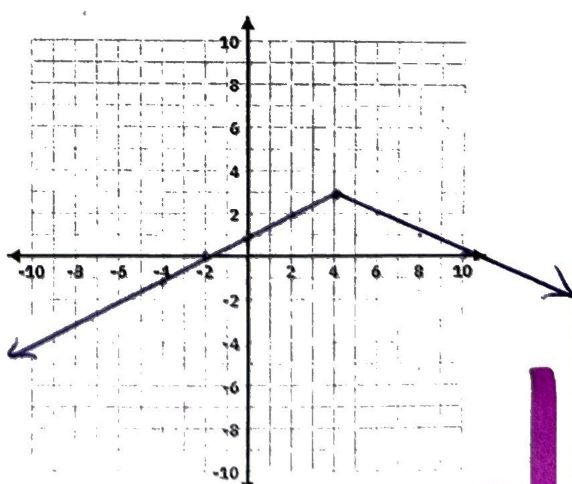
x-intercept(s): $(-2,0)\ (10,0)$ Y-intercept: $(0,1)$

Vertex: $(4, 3)$ Axis of Symmetry: $x = 4$

Max: $(4, 3)$ OR Min: none

Function Notation Description: $-\frac{1}{2}f(x-4)+3$

Verbal Description: Reflect x, VC by $\frac{1}{2}$ Right 4, Up 3



4. Equation: $f(x) = 3|x + 2| - 3$

Domain (interval): $(-\infty, \infty)$ Range (interval): $(-\infty, -3]$

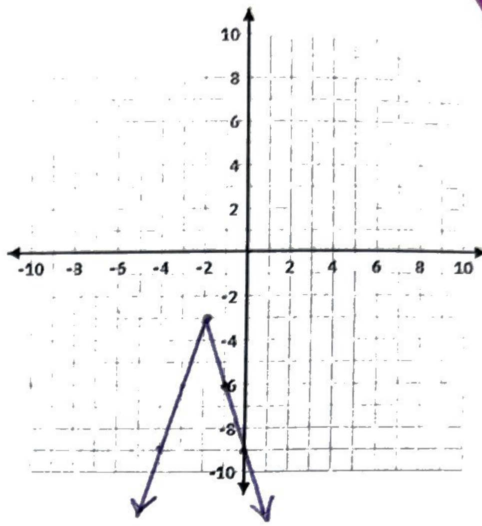
x-intercept(s): none Y-intercept: $(0, -9)$

Vertex: $(-2, -3)$ Axis of Symmetry: $X = -2$

Max: $(-2, -3)$ OR Min: none

Function Notation Description: $3f(x+2) - 3$

Verbal Description: VS by 3, Left 2, Down 3




5. Equation: $f(x) = -\frac{1}{3}|x + 4| - 1$

Domain (interval): $(-\infty, \infty)$ Range (interval): $[-1, \infty)$

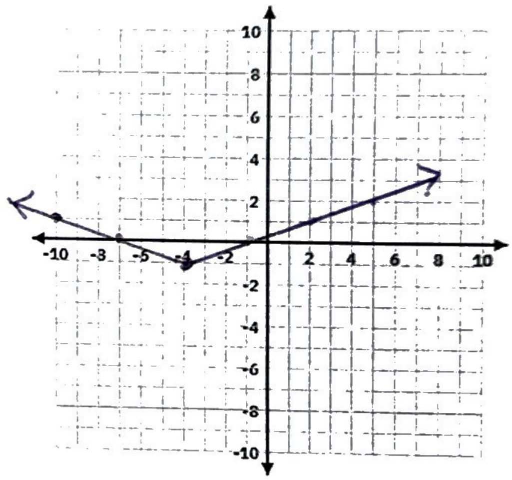
x-intercept(s): $(-7, 0)(-1, 0)$ Y-intercept: $(0, \frac{1}{3})$

Vertex: $(-4, -1)$ Axis of Symmetry: $X = -4$

Max: none OR Min: $(-4, -1)$

Function Notation Description: $-\frac{1}{3}f(x+4) - 1$

Verbal Description: Reflect over x, VC by $\frac{1}{3}$
Left 4, Down 1




6. Equation: $f(x) = -3|x + 2| + 9$

Domain (interval): $(-\infty, \infty)$ Range (interval): $(-\infty, 9]$

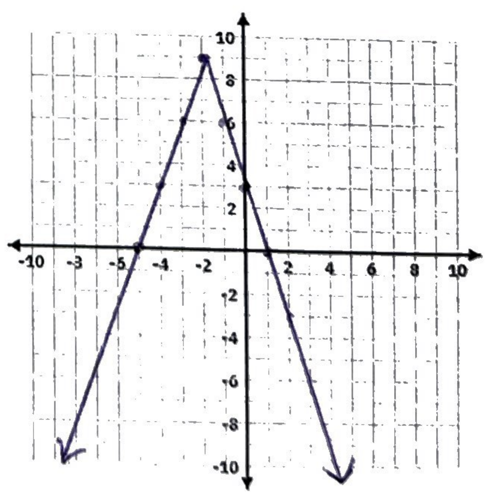
x-intercept(s): $(-5, 0)(1, 0)$ Y-intercept: $(0, 3)$

Vertex: $(-2, 9)$ Axis of Symmetry: $X = -2$

Max: $(-2, 9)$ OR Min: none

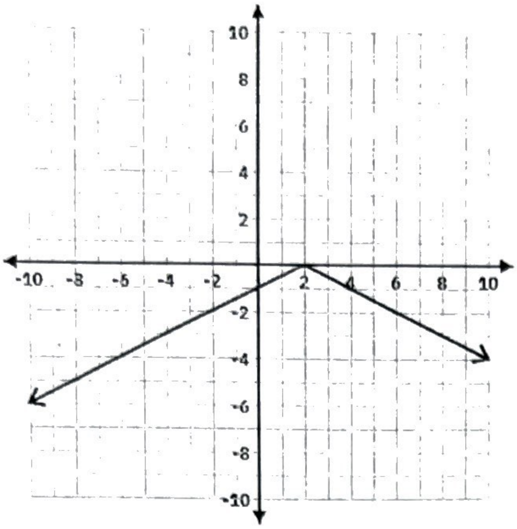
Function Notation Description: $-3f(x+2) + 9$

Verbal Description: Reflect x, VS by 3,
Left 2, Up 9



Given the graph, write the equation, and provide all information.

7.




Equation: $f(x) = -\frac{1}{2}|x-2|$

Domain (interval): $(-\infty, \infty)$ Range (interval): $(-\infty, 0]$

x-intercept(s): $(2,0)$ Y-intercept: $(0,-1)$

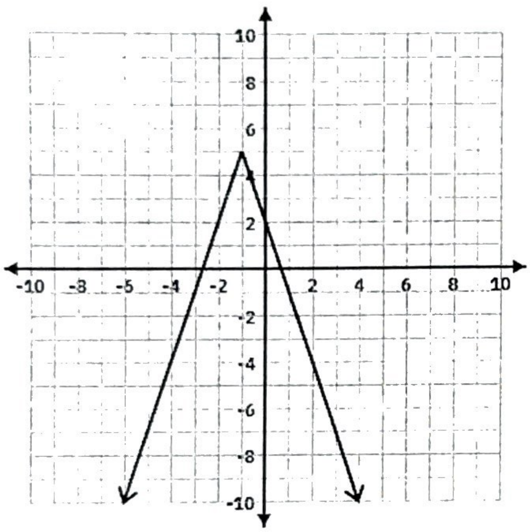
Vertex: $(2,0)$ Axis of Symmetry: $x=2$

Max: $(2,0)$ OR Min: none

Function Notation Description: $-\frac{1}{2}f(x-2)$

Verbal Description: Reflect x, Right 2, VC by $\frac{1}{2}$

8.




Equation: $f(x) = -3|x+1|+5$

Domain (interval): $(-\infty, \infty)$ Range (interval): $(-\infty, 5]$

x-intercept(s): $(-2\frac{2}{3},0)(\frac{2}{3},0)$ Y-intercept: $(0,2)$

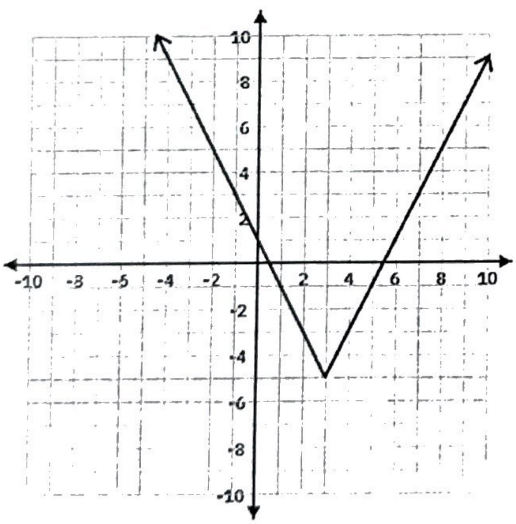
Vertex: $(-1,5)$ Axis of Symmetry: $x=-1$

Max: $(-1,5)$ OR Min: none

Function Notation Description: $-3f(x+1)+5$

Verbal Description: Reflect over x, Left 1, Up 5, VS by 3

9.




Equation: $f(x) = 2|x-3|-5$

Domain (interval): $(-\infty, \infty)$ Range (interval): $[-5, \infty)$

x-intercept(s): $(\frac{1}{2},0)(5\frac{1}{2},0)$ Y-intercept: $(0,1)$

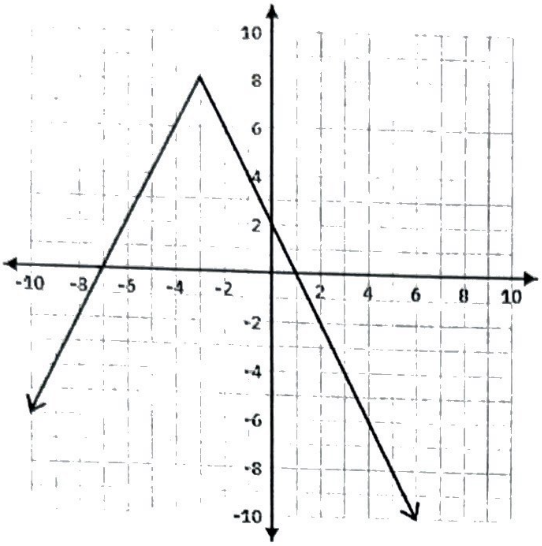
Vertex: $(3,-5)$ Axis of Symmetry: $x=3$

Max: none OR Min: $(3,-5)$

Function Notation Description: $2f(x-3)-5$

Verbal Description: Right 3, Down 5 VS by 2

10.




Equation: $f(x) = -2|x+3|+8$

Domain (interval): $(-\infty, \infty)$ Range (interval): $(-\infty, 8]$

x-intercept(s): $(-7,0)$ $(1,0)$ Y-intercept: $(0,2)$

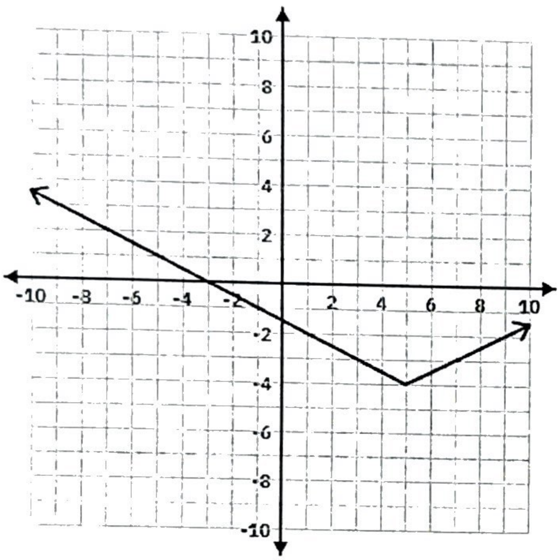
Vertex: $(-3,8)$ Axis of Symmetry: $x=-3$

Max: $(-3,8)$ OR Min: none

Function Notation Description: $-2f(x+3)+8$

Verbal Description: Reflect x, Left 3, Up 8, VS by 2

11.




Equation: $f(x) = \frac{1}{2}|x-5|-4$

Domain (interval): $(-\infty, \infty)$ Range (interval): $[-4, \infty)$

x-intercept(s): $(-3,0)$ $(13,0)$ Y-intercept: $(0,-1.5)$

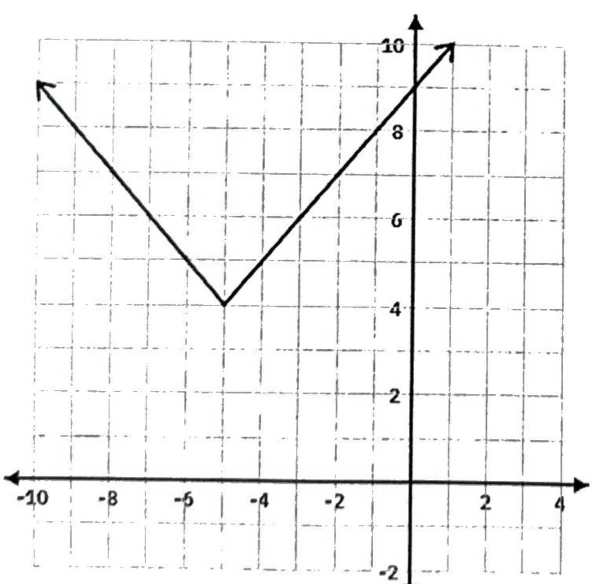
Vertex: $(5,-4)$ Axis of Symmetry: $x=5$

Max: none OR Min: $(5,-4)$

Function Notation Description: $\frac{1}{2}f(x-5)-4$

Verbal Description: Right 5, Down 4 VC by $\frac{1}{2}$

12.




Equation: $f(x) = |x+5|+4$

Domain (interval): $(-\infty, \infty)$ Range (interval): $[4, \infty)$

x-intercept(s): none Y-intercept: $(0,9)$

Vertex: $(-5,4)$ Axis of Symmetry: $x=-5$

Max: none OR Min: $(-5,4)$

Function Notation Description: $f(x+5)+4$

Verbal Description: Left 5, Up 4

**Given a verbal description, write the equation or other requested information.**

13. The function $f(x) = |x|$ is transformed to form a new function $g(x)$, which contains the following: Vertical shift 2 units down, Horizontal shift 3 units to the left

What is the equation for $g(x)$? $g(x) = |x+3| - 2$

14. The function $f(x) = |x|$ is transformed to form a new function $g(x)$, which contains the following: Reflect over x-axis, Vertical shift 4 unit up, Vertical Stretch by 7.

What is the equation for $g(x)$? $g(x) = -7|x| + 4$

15. The function $f(x) = |x|$ is transformed to form a new function $g(x)$, which contains the following: Reflect over y-axis, Vertical shift 6 unit up, Vertical compression by 1/2.

What is the equation for $g(x)$? $g(x) = \frac{1}{2}|-x| + 6$

16. The function $f(x) = |x|$ is transformed to form a new function $g(x)$, which contains the following: Vertical stretch by 3, Vertical shift 6 units up, Horizontal shift 6 units to the right.

What is the equation for $g(x)$? $g(x) = 3|x-6| + 6$

17. The function $f(x) = |x|$ is transformed to form a new function $g(x)$, which contains the following: Reflect over x-axis, Vertical compression by 1/3, Vertical shift 1 unit down, Horizontal shift 5 units to the right.

What is the equation for $g(x)$? $g(x) = -\frac{1}{3}|x-5| - 1$

18. If an absolute value function as a domain of all real numbers and x-intercepts (-4, 0) and (-2, 0) with a vertical stretch of 2,

What is the vertex? (-3,-2)

Don't Worry about 18-25